Surge Protection Devices (SPD)

How they work, 3P vs 4P vs 3+1, and when we need them

What it is

A voltage-triggered switch. It does nothing until a surge arrives, then sends the surge to earth in nanoseconds.

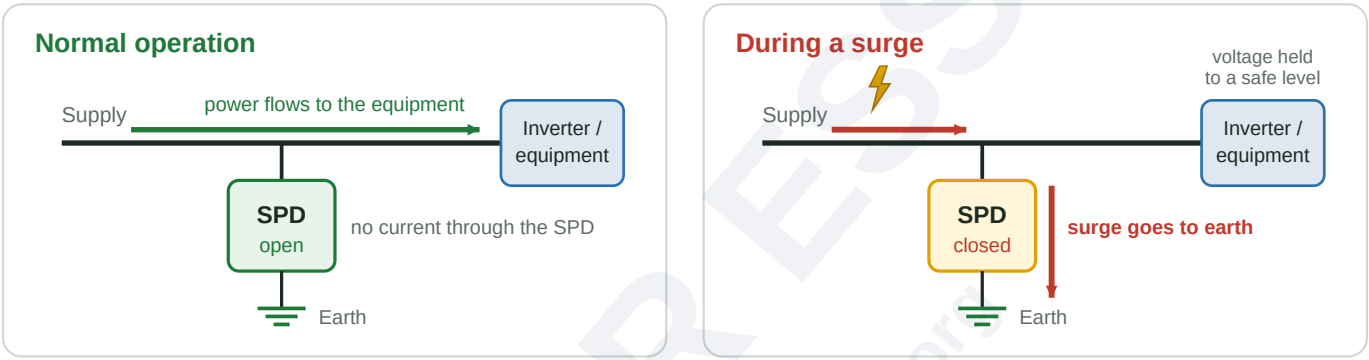
3P or 4P?

Not a quality level. It depends on where the neutral is connected to earth at that board.

Do we need it?

For solar in Thailand: yes, as standard. Only a small house with short cable runs can reasonably go without.

1 How an SPD works



- **Varistor (MOV):** the main part. Very high resistance at normal voltage, very low resistance above its threshold. It clamps the voltage, then recovers.
- **Gas discharge tube (GDT):** a sealed spark gap. Fully open until it fires, so it leaks no current. Used between neutral and earth.
- **It wears out.** Every surge ages the varistor. The window turns from green to red when the module must be replaced.
- **It can fail from overvoltage.** A lost neutral or a utility fault can overheat it, so it needs a thermal disconnector and a backup fuse or breaker.

The 50 cm rule

Keep the wires to and from the SPD under **50 cm in total**. A typical surge adds about 1,000 V for every metre of connecting cable, so a good SPD on long leads protects poorly.

SPD types

Type 1

Direct lightning current. Fitted at the main board when the building has a lightning rod or overhead supply.

Type 2

Induced and switching surges. The standard choice for boards, inverters and DC combiners.

Type 3

Fine protection next to sensitive equipment. Only used together with a Type 2 upstream.

Reading the label

Uc

Highest voltage it can sit on all day. Must be above the supply voltage.

In

Surge current it handles again and again. 20 kA is typical for Type 2.

Imax

Largest single surge it survives. 40 kA is typical for Type 2.

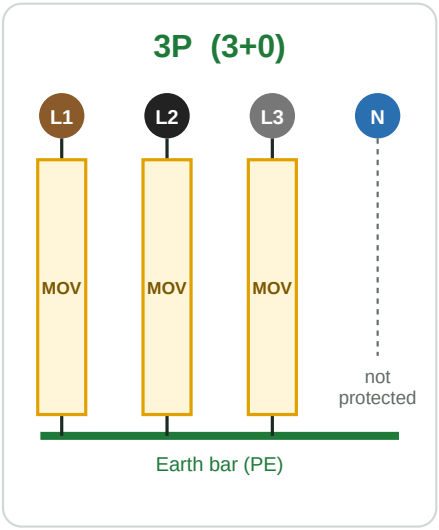
Up

Voltage it lets through. Lower is better: 1.5 kV or less for electronics.

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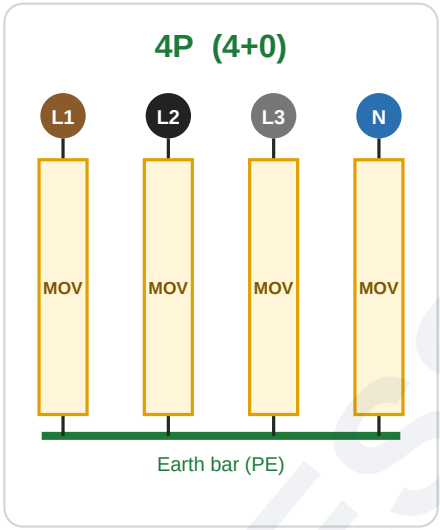
2 3P, 4P and 3+1: what is different

All three protect a three-phase supply. The difference is **which wires are connected to what** inside the device.



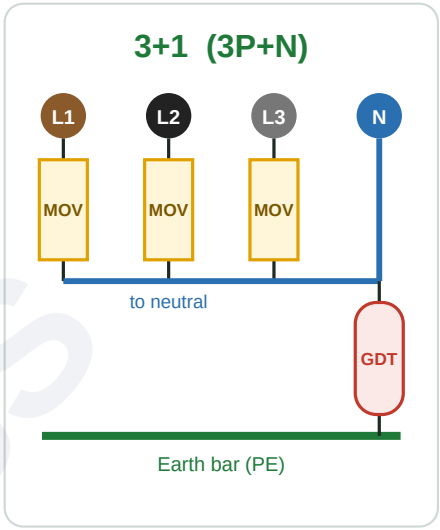
Each phase to earth. Neutral has no pole.

Use at the **main board where neutral and earth are joined** (TN-C, or the main board of a TN-C-S supply), or where there is no neutral.



All four wires to earth through varistors.

Use at **boards after the neutral-earth link**, where neutral and earth run as separate wires (TN-S).



Phases to neutral through varistors, then neutral to earth through a gas tube.

Use where the site has **its own earth rod and no neutral-earth link** (TT), ahead of the RCD.

Quick chooser

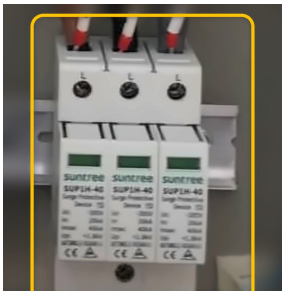
At this board, is neutral joined to earth?

- Yes, joined here → **3P**
- No, separate N and PE bars → **4P**
- No link at all, own earth rod → **3+1**

Why 3+1 exists

On a site with its own earth rod, the path through the ground has high resistance. If a varistor wired phase-to-earth fails short, the fault current is too small to trip the breaker and the earth wire can become live. In a 3+1 device a failed varistor makes a phase-to-neutral short circuit, which trips the breaker normally.

Device types as described in IEC 60364-5-534 (connection types CT1 and CT2). Confirm the earthing arrangement on site before ordering.



Example: a 3P SPD in a main board

Three single-pole modules, one each on L1, L2 and L3, with a common connection to the earth bar below. There is no module on the neutral. This is correct where neutral and earth are joined at this board; on a board with separate neutral and earth bars, a 4P device would be needed.

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3 Do we need it every time?

The general rule (IEC 60364-4-44)

Surge protection is **required** where a surge could cause:

- injury or loss of life
- loss of a public service
- interruption of a business
- harm to many people in one place

Elsewhere a risk calculation decides. A single house may go without if the owner agrees.

The solar rule (IEC 60364-7-712)

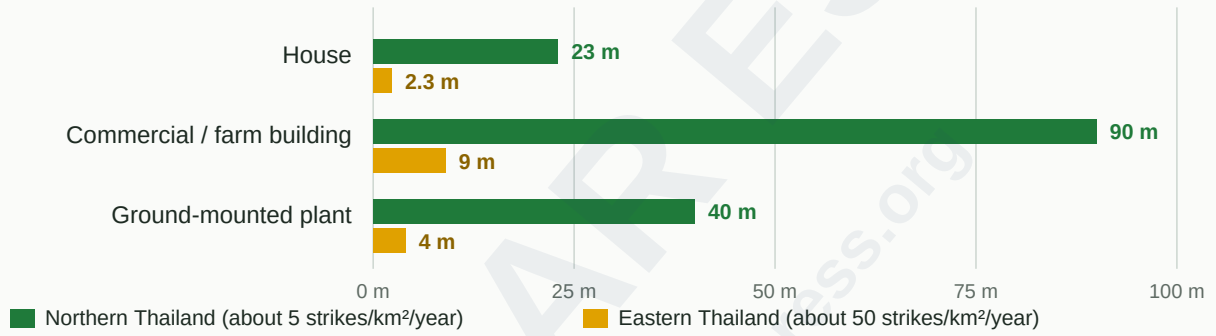
DC side: fit an SPD when the DC cable run is longer than a critical length:

House	**115 ÷ Ng m**
Commercial / farm building	**450 ÷ Ng m**
Ground-mounted plant	**200 ÷ Ng m**

Ng = lightning strikes per km² per year.

AC side: fit an SPD when the inverter is more than **10 m** from the main board.

What that means in Thailand: DC cable length above which an SPD is needed



In the east and south almost every system needs DC surge protection. In the north, a ground mount or carport with a long run to the inverter room does. The northern figure comes from one 2017–2019 study using a global network that misses many strikes, so treat it as a low estimate.

4 Evidence from the field

220,000
lightning and surge insurance claims in Germany in 2024, costing about €350 million (average €1,600 each). Highest in over 20 years.
GDV, German Insurance Association

9.8%
of natural-catastrophe incidents at solar farms are lightning. Average claim: $73,394. Inverters, combiners and bypass diodes are hit most.
Clir, 250+ GW dataset

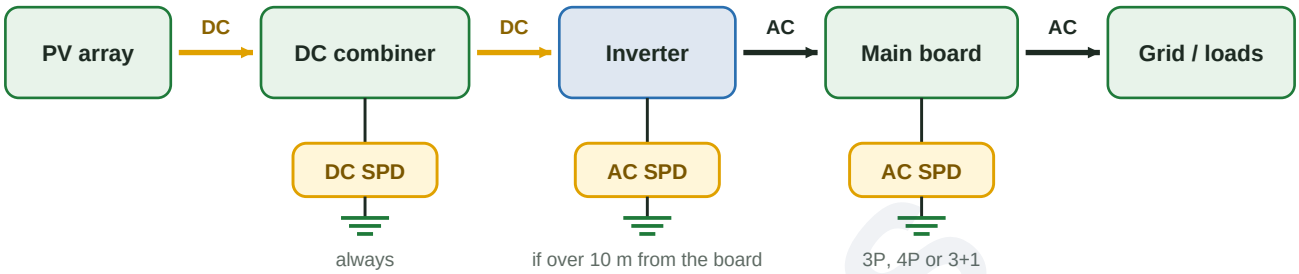
~95%
of SPD failures come from sustained overvoltage (lost neutral, utility fault, wrong wiring), not from lightning itself.
EC&M, 2008

1 arrester
was destroyed at a Greek island solar plant while the electronics behind it survived. Induced surges, not direct strikes, were the likely cause.
NIST field investigation, 1989

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5 Where the SPDs go in a solar system

Add a second DC SPD at the array if the DC run is over 10 m



6 Can we go without?

Can reasonably skip

- A single house with a short DC run (under the critical length), the inverter within 10 m of the main board, an underground supply, and an owner who accepts the risk in writing.
- The second DC SPD at the array end, when the DC run is under 10 m.

Always fit

- Commercial sites, resorts, clinics, farms
- Any system with batteries
- Ground mounts and carports with long cable runs
- Sites fed by overhead lines, or with a lightning rod
- Off-grid and generator sites

7 Solar ESS checklist

- [x] DC SPD (Type 2) in every DC combiner, rated above the string open-circuit voltage.
- [x] AC SPD at the inverter output when it is more than 10 m from the main board.
- [x] AC SPD at the main board: choose 3P, 4P or 3+1 from the earthing arrangement on site.
- [x] Leads under 50 cm in total, with a backup fuse or breaker as the maker specifies.
- [x] Check the indicator windows at each maintenance visit and after every storm season.

Not confirmed in this note

The Thai (EIT / PEA) wording on SPDs for rooftop solar; a lightning-density figure for Chiang Rai specifically; a reliable statistic for the share of inverter failures caused by surges.

Sources

IEC 60364-4-44 clause 443 as summarised by Voltimum (BS 7671) · Schneider Electric Electrical Installation Guide: SPD for photovoltaic applications; Connection of SPD · Electrical Engineering Portal: SPDs and RCDs · DEHN white paper WP018 (2015) · Areerachakul et al., Kalasin University Journal of Science Technology and Innovation (2024) · GDV Blitzbilanz 2024 · Clir: lightning and solar farms · Sandia SAND2000-0876C · NIST, F. D. Martzloff (1989) · EC&M, Mossop and Chiste (2008).

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